

History of Computers

Week 1

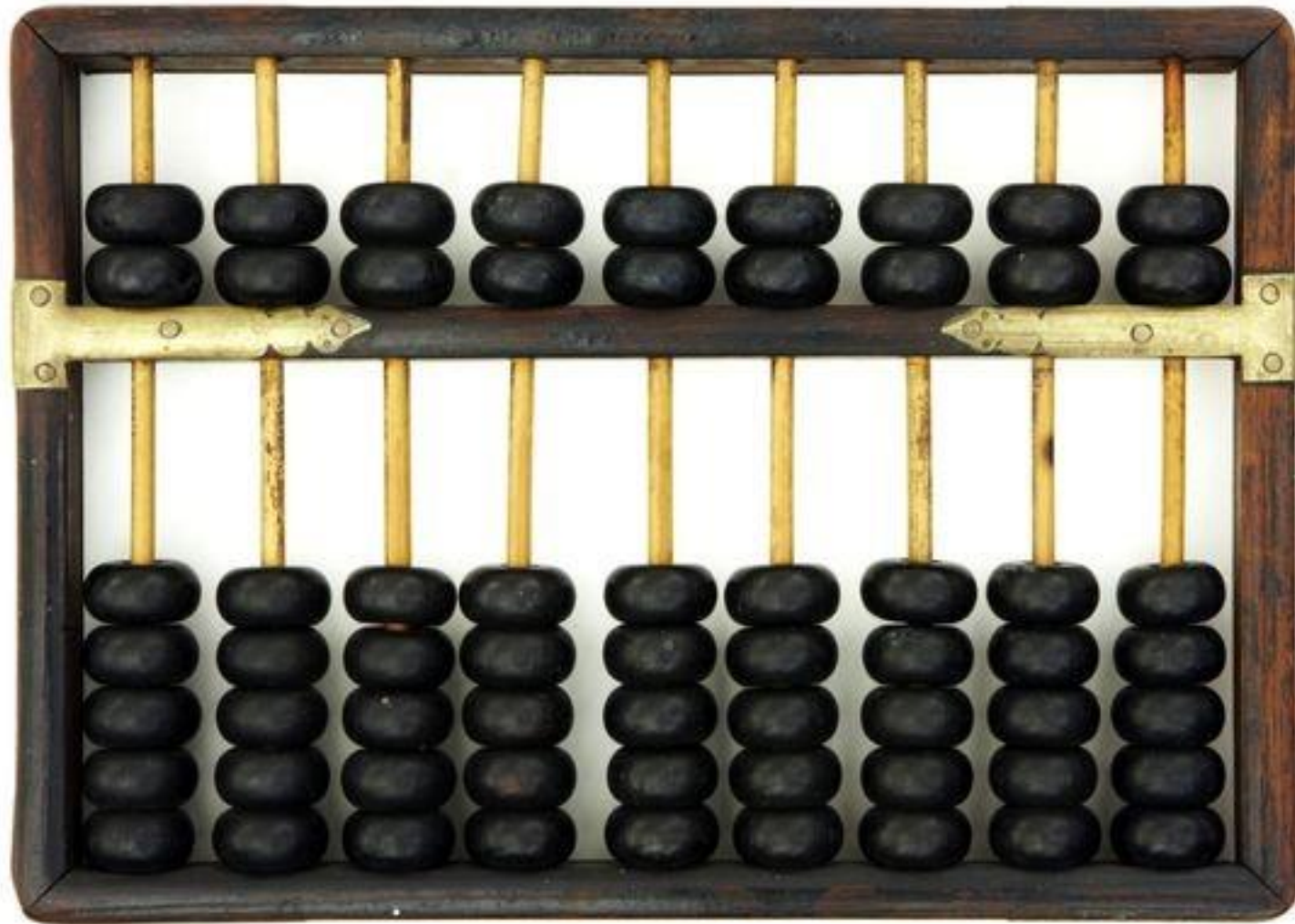
Definition of a Computer

- A computer is defined in the following ways
 - By the work it does
 - By the kind of information it handles
 - By its size and price

Early Calculating Machines

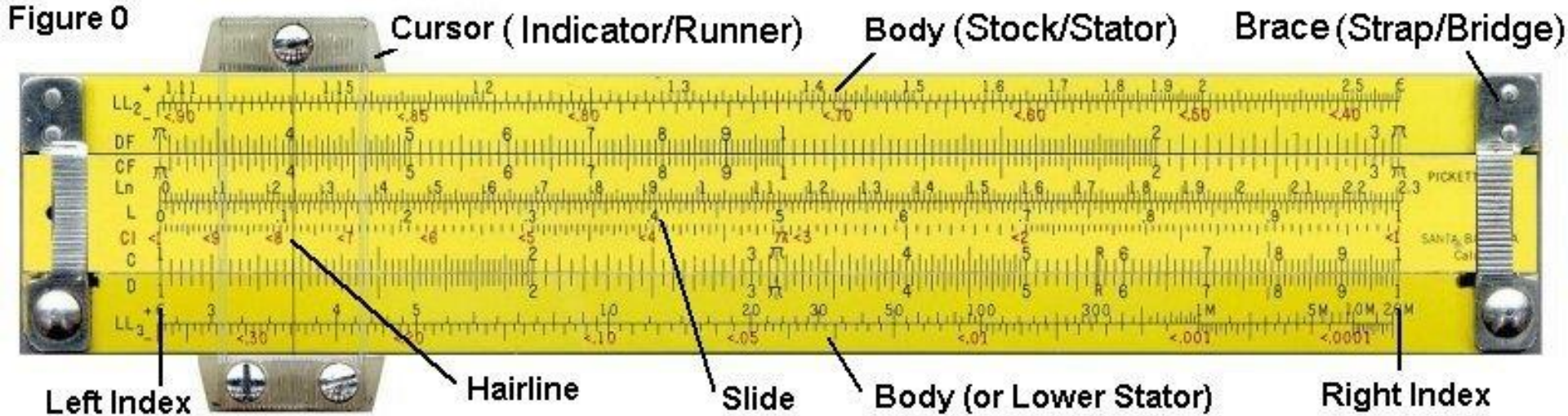
- Abacus
- Slide rule
- Mechanical calculator
- Jacquard machine
 - Built by Jacquard Loom
 - Weaving innovation in the textile industry
 - Punch cards
- Difference engine:
 - Designed by Charles Babbage
 - Automatic mechanical calculator
 - Closed to be the first computer
 - Processed polynomial functions

Abacus



Slider rule

Figure 0



Mechanical Calculator

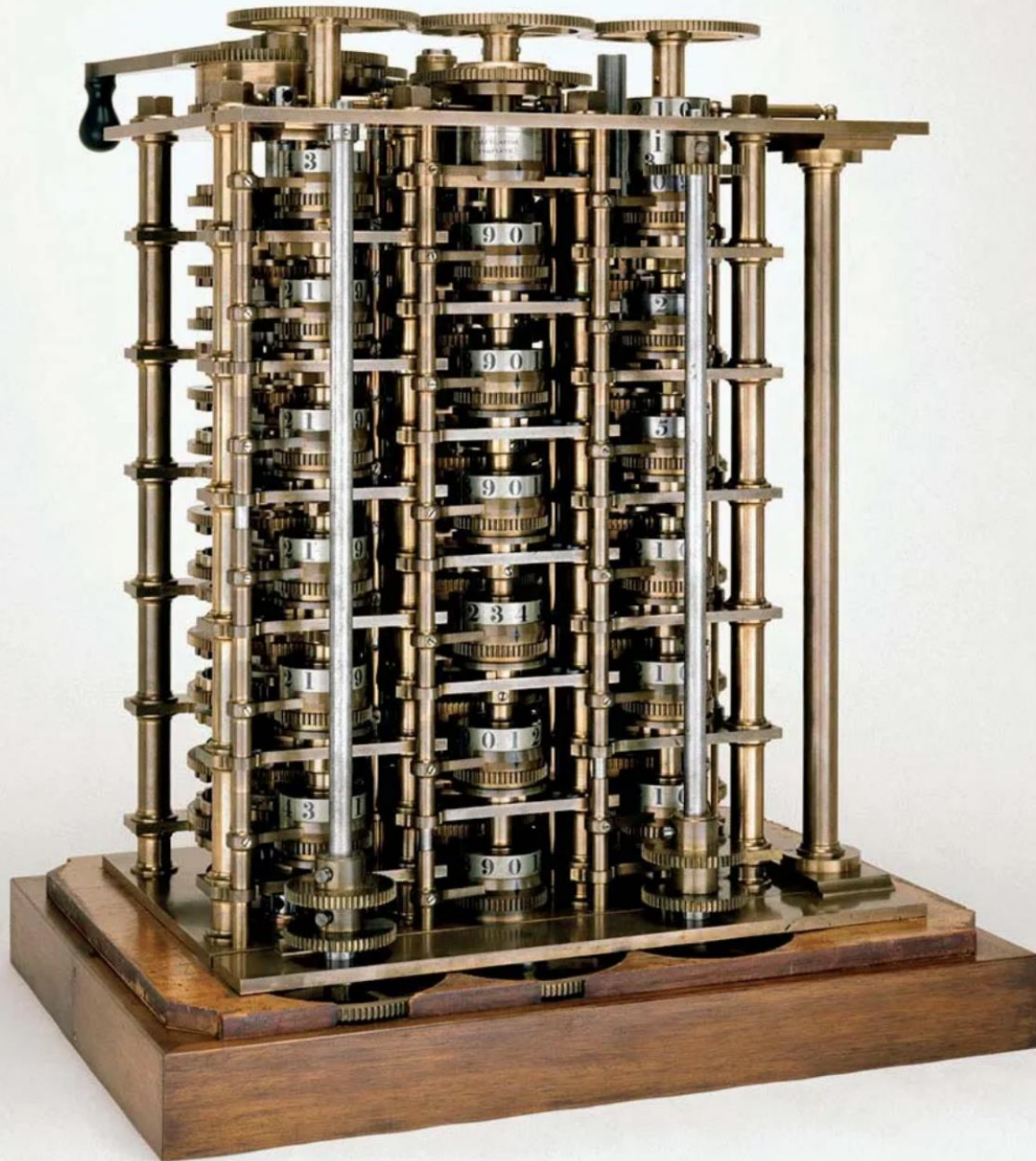


eugene.kaspersky.com

Jacquard Loom's machine



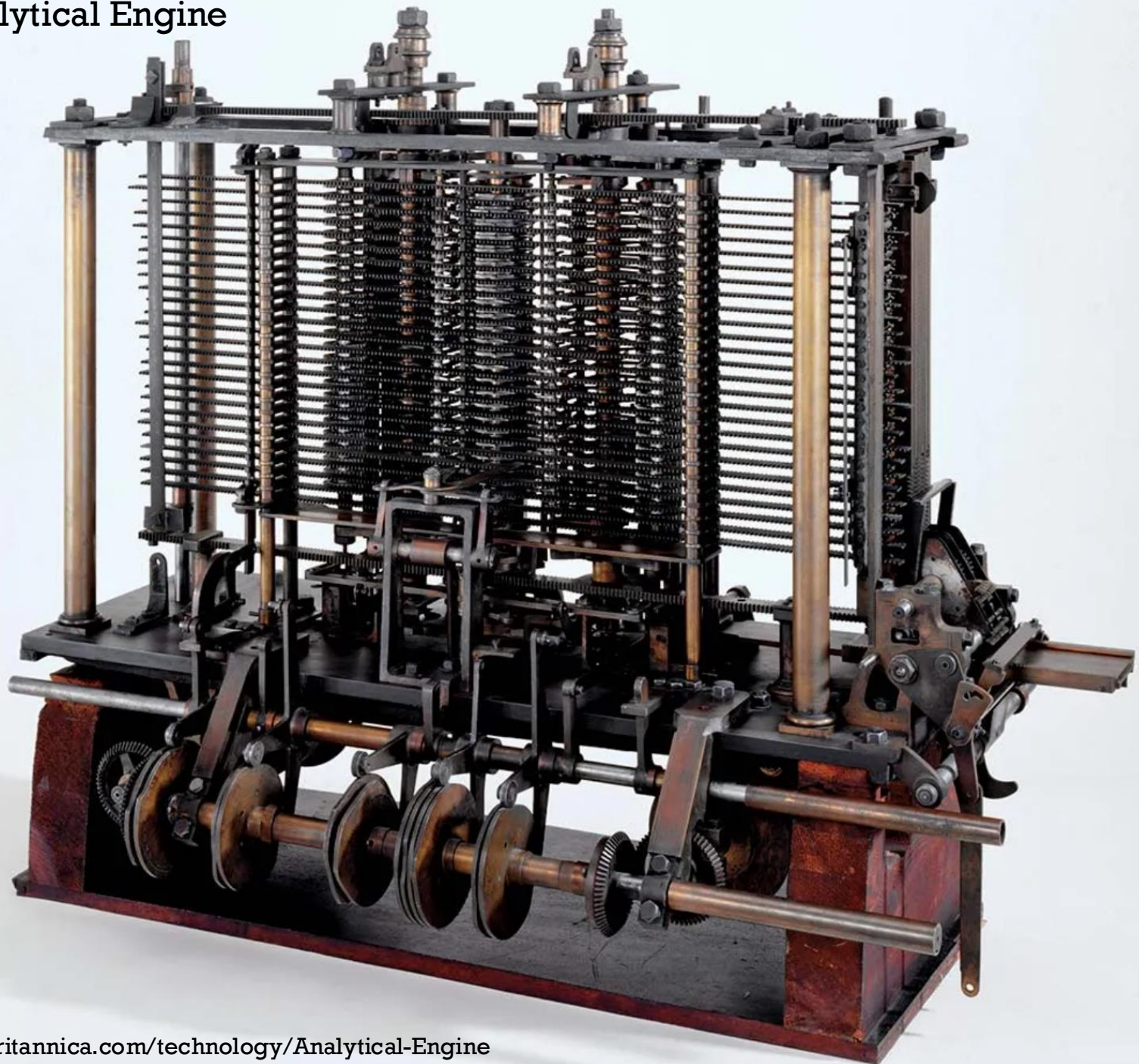
Difference Engine



Early Calculating Machines through those of today

- Analytical engine
 - Designed by Charles Babbage
 - First generally-purposed computer
- The 1890 Census machine
 - Process data for the 1890 U.S. Census
 - Used punched cards to store information
- Electronic Numerical Integrator and Computer (ENIAC)
 - First programmable general-purpose electronic digital computer
 - Used in the construction of a hydrogen bomb
 - More than 17,000 Vacuum tubes, resistors, capacitors and switches
- The transistor
 - Semiconductor device that can amplify or switch electrical signals.
- The Personal Computer (PC)

Analytical Engine

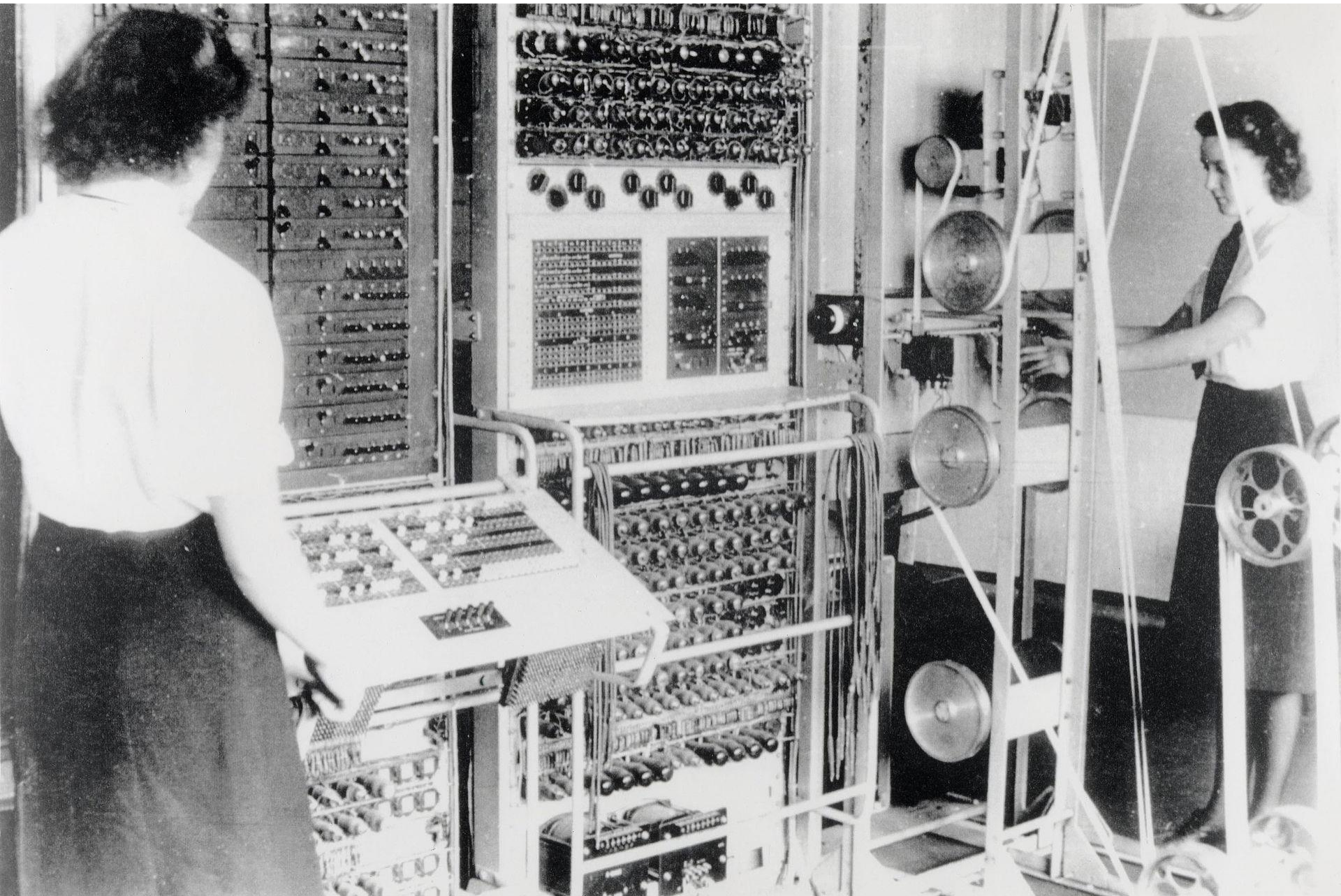


The 1890 Census machine





Colossus Mark 1 and 2



The transistor





Storing Data

- To store data into a variable you can use

1. The assignment operator

```
int x = 15;
```

```
x = 90;
```

2. The standard input stream to read data from the keyboard